

F1 Lap Time Model

Key Findings

2024 Season — Bahrain, Saudi Arabia, Australia, Japan

Feature Importance Breakdown

The model was trained to predict each lap's time relative to that circuit's median — removing circuit-to-circuit baseline differences so it learns what actually drives variation within a race.

What	Share	What it means
Fuel load	22%	Car starts at ~110 kg of fuel. Every lap burns ~1.6 kg. That's 50–60 kg gone by race end — a massive reduction in mass through every corner and braking zone.
Lap number	21%	Closely correlated with fuel burn, but also captures track rubber building up and random safety car timing.
Air temperature	18%	Affects how quickly tyres reach their operating window and how long they stay there. Hot air = tyres overheat faster.
Track temperature	15%	Surface grip changes through the day. A track at 50°C behaves very differently from the same track at 30°C.
Driver	10%	Real pace differences between drivers, and management style — some are harder on tyres, some carry more speed through specific corners.
Tyre compound	6%	SOFT vs MEDIUM vs HARD does produce a measurable baseline pace difference.
Tyre life	3.3%	How many laps are on the current set — the degradation signal itself.
Compound x tyre life	1.8%	Whether the degradation rate differs between compounds (it does, slightly).
Out-lap flag	1.7%	The first lap of every stint is genuinely slower — cold rubber on cold or cooling tarmac.

The Fuel Finding in Plain Terms

A car at the start of the race weighs roughly 200 kg more than at the end when you include fuel. In F1, every 10 kg of additional mass costs approximately 0.3–0.35 seconds per lap. That means fuel load alone can account for 5–7 seconds of lap time improvement across a race — just from the car getting lighter, with nothing else changing.

This is why lap times naturally improve through the first 20–30 laps even when tyres are degrading. The two effects work against each other: fuel getting lighter (faster), tyres wearing (slower). Strategy engineers are constantly balancing both.

The Tyre Finding — 3.3% Is Small but Meaningful

Tyre life at 3.3% sounds minor, but the model is measuring variation within a race, not between races. In absolute terms, a tyre going from lap 1 to lap 30 might add 1–2 seconds per lap by the end of a long stint — an enormous strategic cost. The 3.3% tells you it is a smaller source of lap-to-lap variation than fuel burn, not that it is unimportant.

The compound code at 6% captures the baseline difference: a fresh soft tyre is inherently faster than a hard tyre at the same fuel load and conditions. Teams choose compounds based on that baseline gap versus how long each compound survives — the classic pace vs durability trade-off.

Why This Matters for Race Strategy

The model is essentially saying: if you want to predict a driver's next lap time, knowing their fuel load and which lap of the race it is tells you far more than knowing what compound they are on. Tyre strategy gets the most TV airtime, but the underlying physics of a 200 kg car burning itself lighter across 50+ laps is the bigger story.

Driver vs Car

Separating Driver Skill from Car Performance

2024 Season — Bahrain, Saudi Arabia, Australia, Japan

The Headline Finding: 97% Car, 3% Driver

Of all lap time variation across the 4 races, 96.6% is explained by which car you are in. Only 3.4% comes from the driver. This is a well-known truth in F1 engineering circles but rarely quantified this cleanly.

The methodology is straightforward: teammates share the same car, the same tyres, and race in the same conditions. Any consistent lap time gap between them must come from the driver — not the machinery. Car pace is the average of both teammates, cancelling out driver effects entirely.

This is why driver contracts in F1 hinge so heavily on car competitiveness. A driver in a dominant car will always outscore an equally talented driver in a midfield car. The data makes this unambiguous.

Teammate Gap — Driver vs Driver in Identical Machinery

Sorted by fastest car first. Gaps averaged across all 4 races on normalised lap times.

Team	Faster	Slower	Gap	Car pace vs field median
Red Bull	VER	PER	+0.34s	-1.04s (fastest car)
Ferrari	SAI	LEC	+0.12s	-0.80s
McLaren	NOR	PIA	+0.20s	-0.61s
Mercedes	RUS	HAM	+0.00s	-0.20s (closest pair)
Aston Martin	ALO	STR	+0.27s	+0.16s
Haas	HUL	MAG	+0.55s	+0.53s (largest gap)
Sauber	ZHO	BOT	+0.06s	+0.62s
RB	TSU	RIC	+0.23s	+0.70s
Alpine	OCO	GAS	+0.18s	+0.86s
Williams	ALB	SAR	+0.16s	+0.89s (slowest car)

Notable results

Mercedes — Russell vs Hamilton: +0.004s. Four milliseconds across four races in two of the most experienced drivers on the grid. Statistically identical. This is what elite-level consistency looks like.

Haas — Hulkenberg vs Magnussen: +0.55s. The largest gap on the grid. Hulkenberg is significantly outperforming his machinery; Magnussen is not keeping pace with the car's potential.

Red Bull — Verstappen vs Perez: +0.34s. The most dominant car on the grid (-1.04s vs median), and Verstappen extracts significantly more from it than Perez. Both facts matter for the championship.

Ferrari — Sainz vs Leclerc: +0.12s. One of the tightest midfield gaps — both drivers are close but Sainz edges ahead across these four races.

Car Pace Ranking (Driver Effects Removed)

Car baseline = average of both teammates' median lap times. This is the cleanest measure of actual car performance, independent of who is driving.

1. **Red Bull** -1.04s vs field median — in a class of its own at this point in the season
2. **Ferrari** -0.80s — genuinely competitive but not at Red Bull's level
3. **McLaren** -0.61s — on a strong upward trajectory through 2024
4. **Mercedes** -0.20s — closer to midfield than their reputation suggests
5. **Aston Martin** +0.16s — right at median, strong driver pairing masking car limitations
- 6–10. **Midfield** Haas, Sauber, RB, Alpine, Williams clustered between +0.5s and +0.9s

Why This Matters

The 97/3 split does not mean drivers are unimportant — a 0.5s teammate gap over a season translates to several championship positions and millions in prize money. It means that the single most impactful decision a driver makes is which contract they sign. The best driver in the worst car will still finish behind a competent driver in the best car, virtually every race.

It also explains why mid-season driver swaps rarely fix team performance, why constructor championships often follow from car dominance rather than driver quality, and why the most strategically valuable insight in F1 is accurately knowing where your car actually sits in the pecking order — stripped of the noise that comes from circuit, weather, and driver variation.